Climate-Aware Stochastic Optimization of Hydropower, Renewable Generation, and Battery Storage for Electricity-System Resilience in Albania

Ajsel Budlla · Whitman College, Walla Walla, WA, USA · HYDROFLEX AI — open-source research prototype

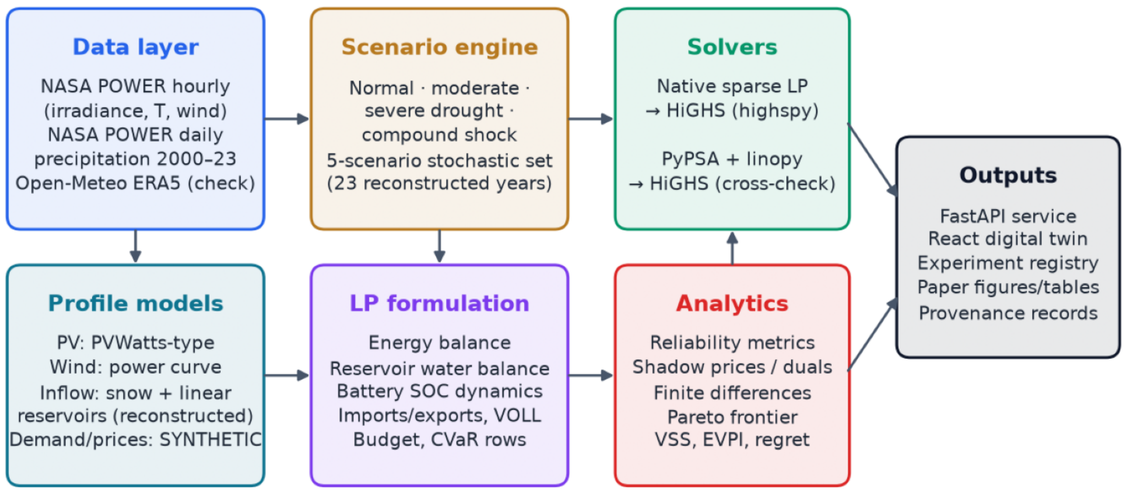
hydroflex-ai.netlify.app · github.com/Ajsel28/hydroflex

Problem & research question

Albania: ≈95% of installed capacity is hydropower; the Drin cascade (Fierza 500, Koman 600, Vau i Dejës 250 MW) dominates supply. Dry years mean imports; IRENA projects ≈15–20% lower average hydro output by 2050.

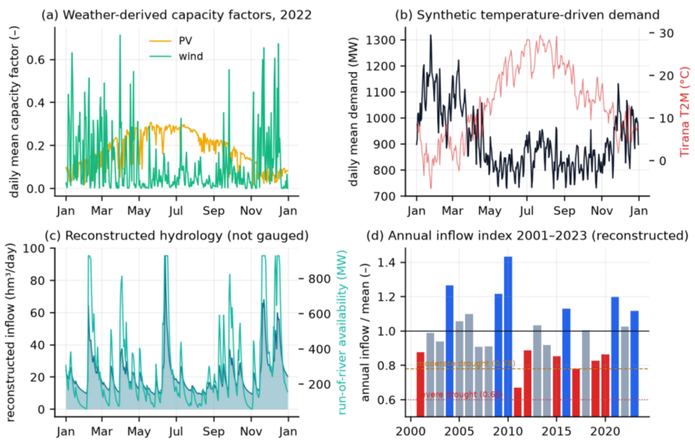
How should a hydropower-dependent system coordinate reservoir operation, solar, wind, batteries and imports under uncertain rainfall, demand and budgets?

Model architecture & method



Linear programme (HiGHS): hourly energy balance · reservoir water balance in hm³ (κ = ρgHη ≈ 552 MWh/hm³) · battery SOC with independent MW/MWh sizing · aggregate interconnector · VOLL-priced shortage · capital budget.
Two-stage stochastic programme: shared capacities, 5 inflow/market scenarios × 8 hourly representative weeks with water-value seasonal boundaries (validated vs full-year chronology); CVaR via Rockafellar–Uryasev. Verified against PyPSA.

Data & scenarios

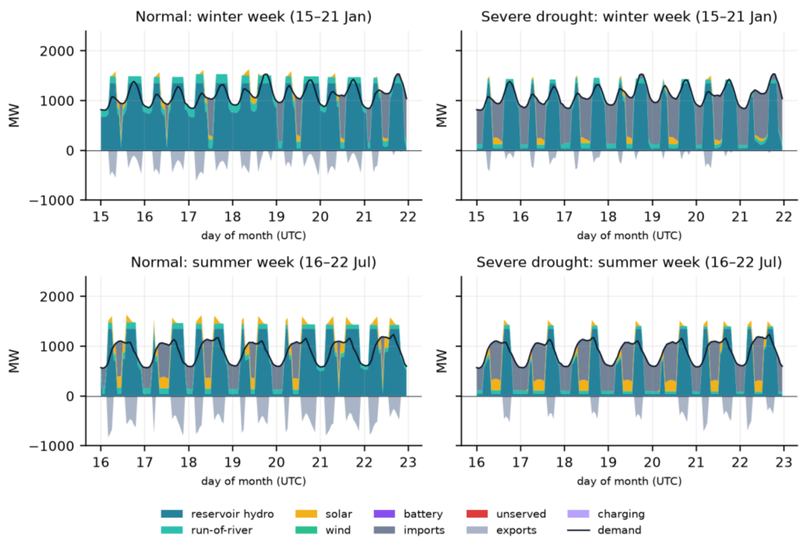


NASA POWER hourly (PV, wind, temperature) and daily basin precipitation 2000–2023 → reconstructed inflow index (5th pct 0.78; min 0.67).

Scenarios: normal · moderate drought (−22%) · severe (−40%) · compound shock.

Demand & prices: SYNTHETIC — not observed grid operations.

Dispatch under drought



Main verified findings

B. Hydropower reduction under drought [supported]

Net imports rise from 0.77 TWh (normal) to 2.26 (moderate) and 3.48 TWh (severe); operating cost rises from 40.8 to 192.7 and 324.5 MEUR/yr. ENS: 0 and 0 MWh.

F. Import price shock and interconnector derating [supported]

Annual cost rises from 324 to 968 MEUR/yr as import prices triple, with ENS 0 MWh. Unserved energy appears at NTC availability ≤ 50% (310.7 GWh).

C. Solar without batteries [supported]

Gross import reduction per added MW: normal +500 MW: 1074 MWh/MW, +1000 MW: 898 MWh/MW, +1500 MW: 699 MWh/MW; severe +500 MW: 1406 MWh/MW, +1000 MW: 1253 MWh/MW, +1500 MW: 1022 MWh/MW. PV curtailment at +1500 MW: normal 21 GWh, severe 2 GWh.

D. Batteries without new renewables [supported]

Annual operating saving (MEUR/yr) normal: 800 MWh: 5.5, 1600 MWh: 9.8, 3200 MWh: 13.8; severe: 800 MWh: 7.0, 1600 MWh: 12.7, 3200 MWh: 17.9. Annualised cost (MEUR/yr): 800 MWh: 25.8, 1600 MWh: 51.7, 3200 MWh: 103.3.

I. Deterministic versus stochastic planning [not supported]

Risk-adjusted VSS = 0.0 MEUR/yr. The stochastic plan costs +0.0 MEUR/yr in expected cost (price of risk aversion) and lowers CVaR(0.9) of operating cost from 397.3 to 397.3 MEUR/yr.

I2. When stochastic planning changes the decision [supported]

With export capacity 300 MW, budget 1500 MEUR and λ=1 the stochastic plan adds pv 1520, wind 230, bp 30, be 66 vs deterministic pv 1232, wind 0, bp 0, be 0; risk-adjusted VSS 71.5 MEUR/yr; CVaR(0.9) 292 → 213 MEUR/yr for +7.8 MEUR/yr expected cost. Full-year chronology: CVaR 282.7 → 208.2 MEUR/yr for +15.1 MEUR/yr expected cost.

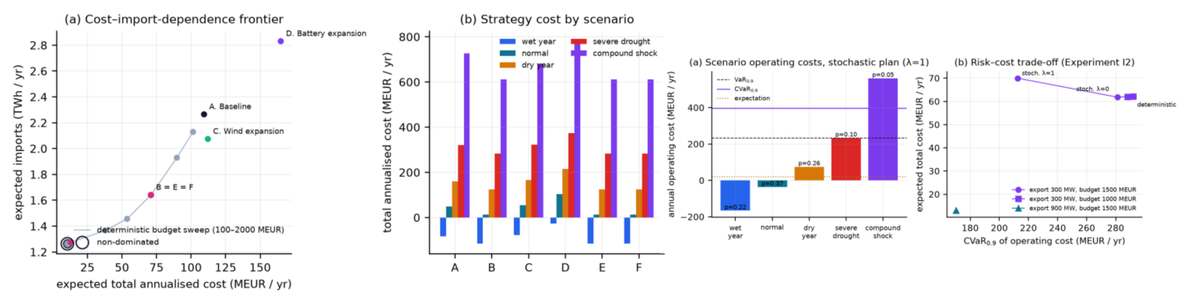
R. Temporal-resolution sensitivity [supported]

Battery operating saving (MEUR/yr): 1 h: 9.8, 2 h: 8.8, 3 h: 7.0, 6 h: 1.3.

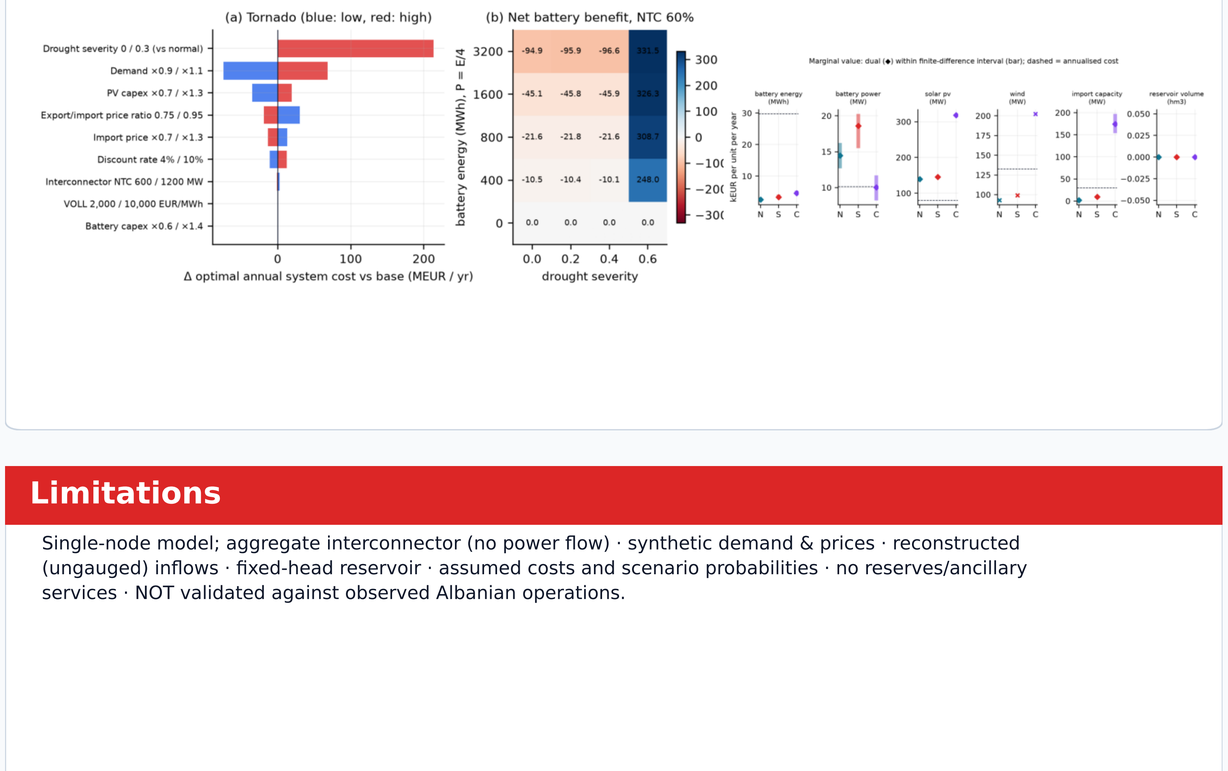
K. Representative weeks vs full-year chronology [supported]

Absolute difference in expected total cost: 1.7–10.0 MEUR/yr. Ranking (chronological): S_budget_1500, F_robust, R_l0.0_b500, B_solar, A_baseline, C_wind, D_battery; (representative weeks): S_budget_1500, B_solar, F_robust, R_l0.0_b500, A_baseline, C_wind, D_battery.

Investment frontier & tail risk



Sensitivity & marginal value of flexibility



Limitations

Single-node model; aggregate interconnector (no power flow) · synthetic demand & prices · reconstructed (ungauged) inflows · fixed-head reservoir · assumed costs and scenario probabilities · no reserves/ancillary services · NOT validated against observed Albanian operations.

Software: Python, HiGHS (highspy), PyPSA/linopy, FastAPI, React. Data: NASA POWER, Open-Meteo ERA5 (cross-check), Natural Earth; costs IRENA Renewable Power Generation Costs in 2024. All results regenerated by `python -m hydroflex.experiments.run_all`. No institutional endorsement is implied.